Do Curved Displays Make for a More Pleasant Experience?

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ABSTRACT

This study investigated the benefits of a curved display compared to a flat display and proposed the optimal radius of curvature for a monitor. The study was carried out in two steps. For identifying the optimal radius, a bendable monitor prototype was used to enable subjects to adjust the display radius manually. Each subject was instructed to search for an optimal radius according to individual preference and visual comfort. Six different themes were applied for the display content. The subjects also reported the radius in which a visual distortion occurred. As a result, it was found that curvature with a radius equal to 600 mm to 700 mm is optimal for a 23-inch diagonal display, while 700 mm to 800 mm is appropriate for a 27-inch diagonal display. Moreover, when the radius of curvature was smaller than 600 mm, a majority reported distortion regardless of the display size. Next, a validation test confirmed that the subjects read the texts faster on the curved display than on the flat display. Based on the empirical results of two experiments, the excellence of a curved monitor in terms of visual comfort, preference, and immersion was verified.

Keywords: curved display, curved monitor, optimal display curvature, visual comfort, preference

1. INTRODUCTION

A curved display recently introduced to the market arouses consumers' interest in terms of the unconventional shape and functional superiority. At certain viewing positions, the curved display provides a constant viewing distance between the eyes and the display.[1] It thus offers users greater visual comfort and a more immersive experience in contrast to the flat display.[2] Currently, the curved display is mainly used for large-size televisions, but in the near future, it is expected to expand into various types of visual display terminals (VDTs). In the use of computer monitors, visual comfort and immersion are especially beneficial because many office workers stay on the computer all day. In observations of 30 workers using multiple monitor configurations in the work place, it was found that all the workers arranged their monitors in an arc shape, which supports comfortable viewing (see Figure 1), rather than a flat shape. This suggests the need for a curved monitor that would enable a single monitor to keep the aforementioned advantages of arch-shaped monitor arrangement.

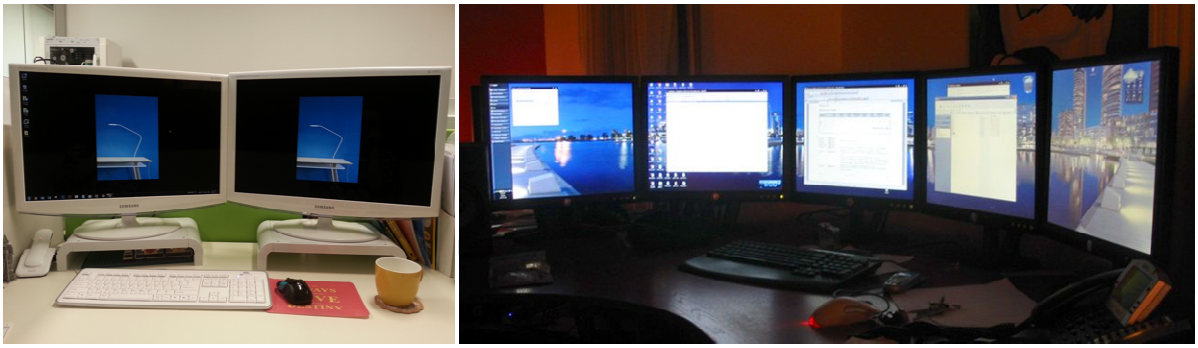

Figure 1. Multiple monitors arranged in an arc shape.

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2. OBJECTIVE

Some previous studies tried to confirm the excellence of a large-size curved display by connecting multiple displays in an arc shape,[3] but no research has yet been performed on a monitor-size display. In this study, a series of user experiments was conducted to examine the benefits of a curved monitor in comparison with a flat monitor. The optimal radius of curvature to provide consumer satisfaction was also investigated in terms of visual comfort and subjective preference.

3. PART I: DISCOVERING THE OPTIMAL RADIUS OF CURVATURE FOR MONITORS

3.1 Objective

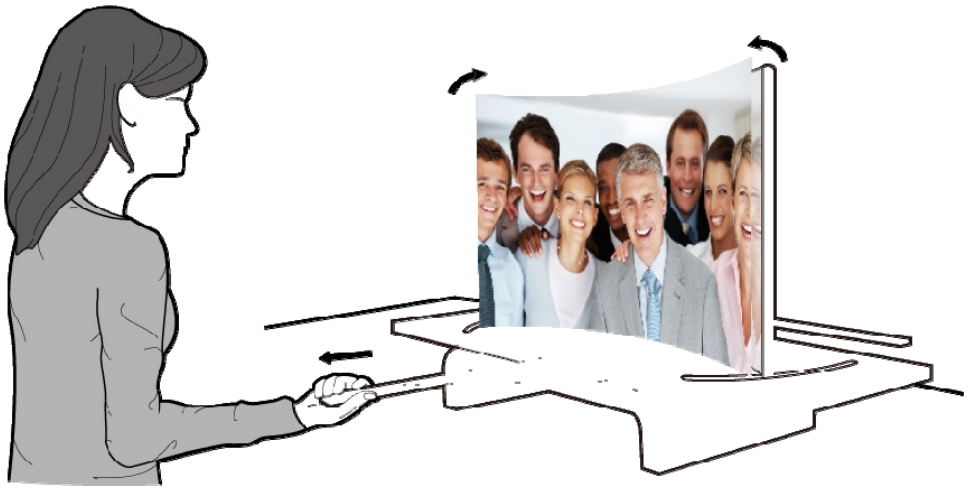
The goal of the experiment in Part I is to discover the optimal radius of curvature for monitor displays to enhance user's visual comfort and subjective preference without leading to visual distortion of the displayed contents.

3.2 Subjects

A total of 50 students comprised of 25 males and 25 females took part in the experiment. The average age of the subjects was 21.84 years with a standard deviation of 2.83 years. All subjects were paid volunteers.

3.3 Stimuli

A bendable monitor prototype was created out of translucent acrylic and paper, as illustrated in Video 1. It allows a subject to manipulate the curvature of the display by pulling or pushing a bar handle, creating a range from a flat display to a curvature with a radius equal to 2,000 mm. The radius increases (close to flat display) if a subject pushes the handle, whereas it decreases (close to convex shape) when a subject pulls the handle. The monitor size was 23-inch in diagonal, and the aspect ratio was set as 16:9.



Video 1. Bendable monitor prototype used in the experiment: the radius of curvature changes by pushing or pulling a handle of the prototype. http://dx.doi.org/10.1117/12.2078102.1

In addition, six content themes, which are widely used in computers, were chosen to simulate the computer monitor. To eliminate the influence of the displayed contents on the subjects' evaluation, the contents consist of (a) plain text, (b) geometrical figures, (c) a worksheet (Microsoft Excel), (d) a photograph of people, (e) wallpaper on a desktop, and, (f) a

3D game image, as shown in Figure 2. All image contents were printed on photo quality paper with two sizes, 23-inch and 27-inch diagonal, and the contents paper can easily be attached and detached on the bendable monitor prototype.

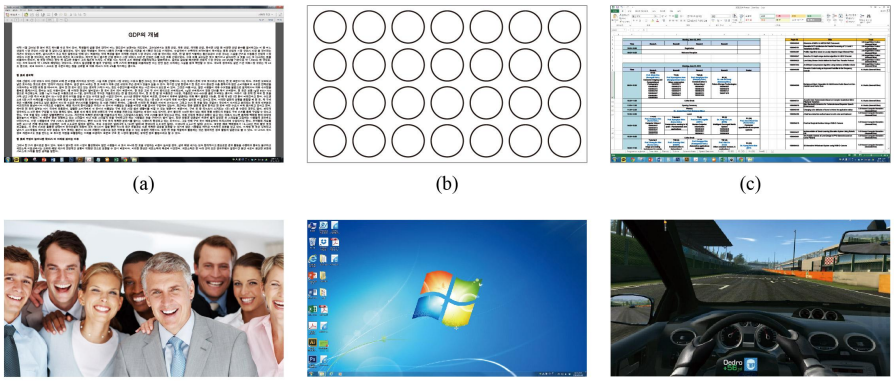


Figure 2. Six content themes displayed on the bendable monitor prototype: (a) plain text, (b) geometrical figures, (c) a worksheet (Microsoft Excel), (d) a photograph of people, (e) wallpaper on a desktop, and (f) a 3D game image.

3.4 Method

The subjects were seated 60 cm from the bendable monitor prototype, a typical viewing distance from the computer monitor.[4] Prior to the experiment, the instructor explained the procedure to the subjects and offered them the opportunity to ask any questions related to the procedure. They then received a five-minute training period to achieve a sufficient ability to control the radius of the curvature.

The subjects were asked to find the optimal radius of curvature for monitor displays by adjusting the bar of the prototype. They were each exposed to the six themes and to two different display sizes, totaling 12 judgments made by each subject. The optimal curvature criteria requested of the subjects included individual preference and visual comfort. In addition, they were requested to define the radius at which a distortion occurred.

At the subject's place, the measure of the correlated color temperature was 6,500 K, and the illuminance was 600 lx. The time taken to complete the experiment was approximately 20 minutes.

3.5 Results and Analysis

The optimal radius of curvature for monitor displays was discovered through the experiment. One-way analysis of variance (ANOVA) was performed using SPSS statistical analysis software to examine the effect of displayed contents on the optimal radius of curvature. The analysis yielded statistical significance at an alpha level of 0.05, thus reporting that the effect was not statistically significant ($p = 0.23$). In other words, the optimal radius of curvature is not affected by displayed contents. Thus, the evaluation results of the six contents have been combined and described as histograms represented by radius of curvature in the horizontal axis and frequency in the vertical axis.

The results of the 23-inch display indicated that the curvature with a radius equal to 600 mm to 700 mm appeared to be perceived dominantly as the optimal curvature, as shown schematically in Figure 3 above, whereas the displays with less than 400 mm curvature or more than 1,300 mm curvature were hardly selected. In terms of the 27-inch display, the curvature with a radius equal to 700 mm to 800 mm received the most favorable evaluation, followed by the curvature with a range between 600 mm and 700 mm, as illustrated in Figure 3 below. The optimal curvature for the 27-inch display is slightly larger than that for the 23-inch display, since the curved form is more recognizable when the display gets larger. Similar to the results of the 23-inch display, there was a sharp decline in the selection frequency in the displays with less than 400 mm curvature or over 1,500 mm curvature.[5]

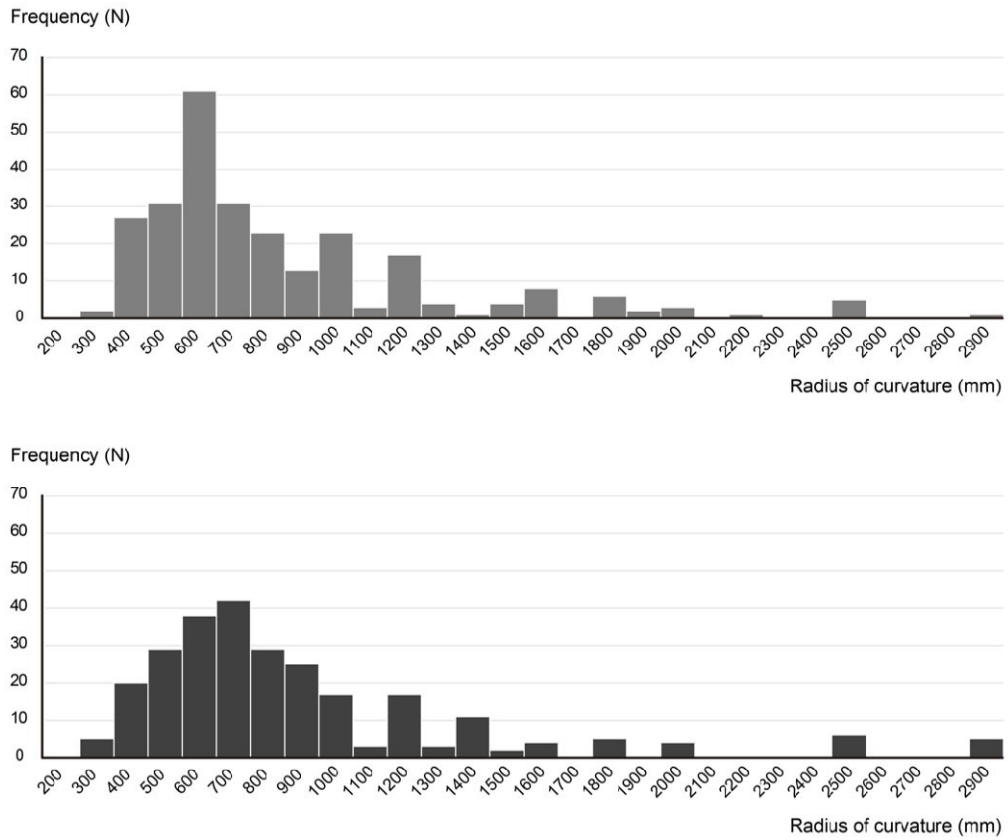


Figure 3. The frequency of the optimal radius of curvature for the 23-inch display (above) and for the 27-inch display (below).

Also, more than 85 percent of subjects reported a visual distortion of the displayed contents when the radius of curvature was smaller than 600 mm regardless of the display size, whereas in the case of 1,000 mm or larger displays they could not see any distortion, as seen in Figure 4. This means that a display with less than 600 mm curvature is inappropriate for a monitor display.

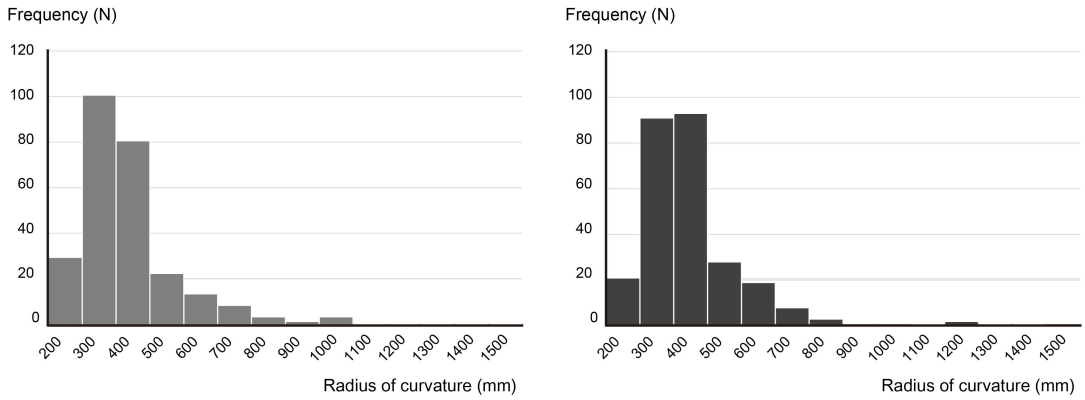


Figure 4. The frequency on the radius of curvature at which a distortion occurred: for 23-inch display (left) and for 27-inch display (right)

By taking two evaluations for finding the optimal radius of curvature and the radius at which a distortion occurred, it was revealed that the curvature with a radius equal to 600 mm to 700 mm is optimal for a 23-inch diagonal display, while the curvature with a radius equal to 700 mm to 800 mm is appropriate for a 27-inch diagonal display.

4. PART II: VALIDATING THE BENEFITS OF CURVED MONITORS

4.1 Objective

In order to provide a more robust evidence of a user test, it is always worth examining the effect by means of different measurements. For example, Na and Suk showed the coherent evidences from people's subjective judgments as well as their brain activities.[6] Through inspecting both psychological and cognitive aspects, we expected to confirm the result in a more objective way. In this foregoing, the experiment in Part II aims to verify the benefits of curved displays compared to flat displays in terms of the aspect of cognitive efficiency.

4.2 Subjects

Six subjects, one male and five female, participated in the experiment. The subjects ranged in age from 23 to 39 years, with the mean age of 26.83 years and a standard deviation of 5.60 years.

4.3 Stimuli

Four different newspaper articles were printed on photo quality paper with a 23-inch diagonal size. The length of each article was approximately 2,500 letters in Korean, and the article consisted of different content but with the same level of difficulty to circumvent the influence of content type.

4.4 Method

The subjects were instructed to read four sets of approximately 2,500 character newspaper articles at a natural speed on a flat display and a curved display in a random order, resulting in two sets of data for each display. The radius of the curved display was individually defined based on each subject's judgment, and they were given a one minute break between reading sessions.

4.5 Results and Analysis

The mean radius of curvature of the six subjects was equal to 633.33 mm with a standard deviation of 177.95 mm, which is similar to the results from the experiment in Part I. The mean values of reading time on the flat display and the curved display were 175 seconds and 160 seconds, respectively. Five subjects out of six read faster on the curved display than the flat display, while the other subject recorded an equal reading time on the two displays (see Table 1). Consequently, the validation test confirmed the assumption that a curved display allows for a better cognitive performance compared to the flat displays, and it is consistent with the findings from the previous study.[5]

Table 1. The mean reading time of each subject on the flat displays and the curved displays.

Display type	**Reading time (s)**						
	subject 1	subject 2	subject 3	subject 4	subject 5	subject 6	**Mean**
Flat display	194	157	221	141	180	157	**175**
Curved display	188	149	216	114	139	157	**160**

5. CONCLUSION

In this study, the optimal radius of curvature for monitor displays to provide consumer satisfaction was examined. The subjects were asked to search for an optimal radius with respect to individual preference and comfort, as well as to report the radius in which a visual distortion occurred. The subjects used a bendable monitor prototype to adjust the display curvature. The results showed that the subjects preferred the display with a range between 600 mm and 700 mm for a 23-inch display, and a display with a range between 700 mm and 800 mm for a 27-inch display. Hence, it is plausible that the optimal radius of curvature varies depending on the display size and may increases as the display becomes larger. Very few people showed a preference for a radius larger than 1,500 mm, and this indicates that the majority of the subjects prefer a curved display than a flat display. Also, most subjects reported a visual distortion when the radius of curvature was smaller than 600 mm, regardless of the display size. The validation test confirmed that the curved displays had better cognitive efficiency compared to the current flat displays. The empirical findings from the study suggest the benefits of curved monitors.

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